Assessment of Healthcare Research in Saudi Arabia Final Version for Reviewer

* Indicates required question

1. What was the PMID of the research paper? *

   Please select the PMID reflected in the Excel sheet provided to you.

2. Do you wish to recuse yourself from reviewing this study for any reason? *

   You should recuse yourself in the following cases:

   - You were an author in the paper.
   - You are a very close friend to one of the authors in the study.
   - You are in the same department as one of the authors in the study.
   - You are a relative to one of the authors in the study.
   - You are a current collaborator with one of the authors in the study.
   - You have a financial conflict of interest with the topic of the paper.

   Mark only one oval.

   ◯ No, I do not wish to recuse myself.
   ◯ Yes, I recuse myself from reviewing this paper. *Skip to question 49*

Study task

3. What was the study task or epidemiological paradigm? *

Please select the task reflected in the Excel sheet provided to you.

Mark only one oval.

- Descriptive *Skip to question 4*
- Predictive *Skip to question 18*
- Causal *Skip to question 20*

Descriptive study design and population

4. What was the study design? *

Mark only one oval.

- Cross-sectional
- Cohort
- Ecological

5. Did the study use data from Saudi Arabia or not? *

Yes: if the data were collected in (or retrieved) in full or part in Saudi Arabia, regardless of participants' nationality.

No: if the data were collected (or retrieved) in full outside of Saudi Arabia.

Mark only one oval.

- Yes
- No

Sampling, sample size, and baseline selection bias due to investigators

6. What was the sampling technique? *

Random:

- If authors conducted any type of random sampling of the target population specified in their methods and (showed how many were invited to the study or showed the response rate).
- In case of secondary data (e.g., registry, EMR/EHR/EDR, claims, etc.) and authors took a random sample of those meeting inclusion/exclusion criteria with specifying the denominator (whole study population that meets inclusion/exclusion criteria.)

Non-random:

- If authors conducted non-probability sample of the target population specified in their methods.
- If authors conducted random sampling **without** stating how many were invited to the study and without stating the response rate.
- If authors included participants **without** mentioning that they screened all medical records.
- If authors only included those with complete data in the study or excluded those with missing data from the study.
- In case of secondary data (e.g., registry, EMR/EHR/EDR, claims, etc.) and authors took a random sample of those meeting inclusion/exclusion criteria **without** specifying the denominator (whole study population that meets inclusion/exclusion criteria.)

Whole population:

- If authors invited everyone in the target population (whole study population that meets inclusion/exclusion criteria) specified in their methods.
- In case of secondary data (e.g., registry, EMR/EHR/EDR, claims, etc.) and authors included all who met inclusion/exclusion criteria (whole study population that meets inclusion/exclusion criteria.)

Mark only one oval.

- Random
- Non-random
- Whole population was invited (no sample was taken) *Skip to question 10*

Sampling, sample size, and baseline selection bias due to investigators

7. Was a sample size calculation done? *

Yes: if authors mentioned in methods.

No: if sampling was done **but no** sample size calculation was done.

Mark only one oval.

- Yes
- No *Skip to question 10*

Sampling, sample size, and baseline selection bias due to investigators

8. Did the authors account for ineligibility, non-response, loss to follow-up, etc. in sample size calculation? *

Yes: If authors took a sample and mentioned that they increased the sample size to address any of these issues.

No: If authors took a sample **but did not** mention that they increased the sample size to address any of these issues.

Mark only one oval.

- Yes
- No

9. Was the sample size needed achieved? *

Yes: if authors reached the minimum sample size suggested by sample size the calculator.

No: if authors took a sample and did a sample size calculation **but did not** reach the sample size determined by the calculation.

Mark only one oval.

- Yes
- No

Sampling, sample size, and baseline selection bias due to investigators

10. Was the baseline selection bias from the investigator accounted for? *

Yes: if authors took a non-random sample (or invited the whole population or took a random sample of participants **but not** everyone agreed to participate) and authors used inverse probability of selection weights.

No: if authors took a non-random sample (or invited the whole population or took a random sample of participants **but not** everyone agreed to participate) and authors **did not** use inverse probability of selection weights.

No baseline selection bias from investigator: if the authors invited the whole population (or took a random sample of participants) and everyone agreed to participate.

Mark only one oval.

- Yes
- No
- No baseline selection bias from investigator

Outcome measurement and its validity

11. What was the nature of the outcome? *

Regardless of how the authors actually measured the variable in the study

Subjective: if there aren't any methods to measure the variable besides relying on the self-report of a participant. These include feelings, pain, quality of life, perception, attitude, etc.

Objective: if there are methods to measure the variable besides relying on the self-report of a participant (e.g., through examination or testing). These include knowledge, body examination, scales, blood tests, body measurements, imaging, etc.

Mark only one oval.

- Subjective
- Objective

12. Did the authors use validated measurement for the outcome? *

First yes: if authors conducted internal validation to calculate sensitivity/specificity/PPV/NPV or cited these numbers (no need to go back and looking at the title and content of the cited paper.)

Second yes: if authors stated that they used a validated measurement without sensitivity/specificity/PPV/NPV or conducted validation with experts (no need to go back and looking at the title and content of the cited paper.)

This type of objective measurement does not usually require validation (e.g., surgical intervention): if authors used an objective measurement that does not usually require validation for logistical reasons such as surgical interventions.

No: if authors **did not** do any criterion/content/face validity themselves or **did not** cite any paper/guideline/classification even if the measurement was considered a usual/gold standard measurement.

Mark only one oval.

- Yes, the authors conducted criterion validity or cited the criterion validity from other studies
- Yes, the authors conducted content/face validity or cited the content/face validity from other studies
- This type of objective measurement does not usually require validation (e.g., surgical intervention)
- No

13. Was the measurement bias in the outcome accounted for? *

(e.g., corrected using internal validation or transported sensitivity, specificity, PPV, or NPV)

Yes: if authors corrected outcome using sens/spec/PPV/NPV.

No: if authors **did not** correct outcome using sens/spec/PPV/NPV.

Mark only one oval.

- Yes
- No

Missing outcome data

14. Was there any participant with a missing outcome? *

If the outcome was measured at multiple time points, then please answer this question based on the main time point of the outcome specified in the methods. If none was specified, then please answer this question based on the last time point.

Yes: If authors stated in the text or tables that at least one participant included in the study had a missing value in outcome.

No: If authors reported all categories of a variable and they sum up to the study population.

Not reported or unknown: If authors **did not** report all categories of a variable (e.g., only reported one category of a binary variable and so on) and did not specifically report how many were missing the outcome in text or tables.

Mark only one oval.

- Yes
- No *Skip to question 16*
- Not reported or unknown *Skip to question 16*

Missing outcome data

15. Any methods used for handling missing data in outcome? *

If the outcome was measured at multiple time points, then please answer this question based on the main time point of the outcome specified in the methods. If none was specified, then please answer this question based on the last time point.

No: if there was missing data in outcome but no imputation was done.

Mark only one oval.

- Yes, missing indicator modeling
- Yes, single imputation with mean value
- Yes, single imputation with predicted value using regression
- Yes, single imputation with last value carried forward
- Yes, multiple imputation
- No

Addressing errors in discussion section

16. Did the authors qualitatively mention the following errors in the discussion section? (select all that apply) *

Please review the whole discussion and not just the limitations section. Only choose an error if it was clearly discussed. If authors vaguely talked about variables and mentioned that they could impact the results without how pointing the type of error it can lead to, please do not select an error based on your interpretation.

Check all that apply.

- [ ] Random error
- [ ] Selection bias
- [ ] Measurement bias
- [ ] Missing data
- [ ] None of the above

Conflating the study task

17. Did the authors investigate any association beyond their primary intention of describing the sample? *

Yes: if authors stated that they were looking for any association between any variables with the study outcome.

No: otherwise.

Mark only one oval.

- Yes
- No

Skip to question 49

Predictive study design and population

18. What was the study design? *

Diagnostic: if authors predicted the probability of having an outcome at present based on other variables.

Prognostic: if authors predicted the probability of having an outcome in the future (or disease progression) based on other variables.

Mark only one oval.

- Diagnostic
- Prognostic

19. Did the study use data from Saudi Arabia or not? *

Yes: if the data were collected (or retrieved) in full or part in Saudi Arabia, regardless of participants' nationality.

No: if the data were collected (or retrieved) in full outside of Saudi Arabia.

Mark only one oval.

- Yes
- No

Skip to question 49

Causal study design and population

20. What was the study design? *

Mark only one oval.

- Cross-sectional
- Case-control
- Cohort (including clinical trials without randomization)
- Ecological
- Case-crossover
- Pre/post
- Quasi-experimental (including instrumental variables, difference-in-difference, regression discontinuity design, and interrupted time series)
- Randomized clinical trial (including all types of randomized experiments)

21. Did the study use data from Saudi Arabia or not? *

Yes: if the data were collected (or retrieved) in full or part in Saudi Arabia, regardless of participants' nationality.

No: if the data were collected (or retrieved) in full outside of Saudi Arabia.

Mark only one oval.

- Yes
- No

Sampling, sample size, and selection bias

22. What was the sampling technique? *

Random:

- If authors conducted any type of random sampling of the target population specified in their methods and (showed how many were invited to the study or showed the response rate).
- In case one group was included as a whole and the other group was randomly sampled (with or without matching.)
- In case of secondary data (e.g., registry, EMR/EHR/EDR, claims, etc.) and authors took a random sample of those meeting inclusion/exclusion criteria with specifying the denominator (whole study population that meets inclusion/exclusion criteria.)

Non-random:

- If authors conducted non-probability sample of the target population specified in their methods.
- If authors conducted random sampling **without** stating how many were invited to the study and without stating the response rate.
- If authors included participants **without** mentioning that they screened all medical records.
- In case one group was included as a whole and the other group was **not** randomly sampled (with or without matching.)
- If authors only included those with complete data in the study or excluded those with missing data from the study.
- In case of secondary data (e.g., registry, EMR/EHR/EDR, claims, etc.) and authors took a random sample of those meeting inclusion/exclusion criteria **without** specifying the denominator (whole study population that meets inclusion/exclusion criteria.)
- In case of case-control study, If authors selected controls from a different source population that gave rise to cases.

Whole population:

- If authors invited everyone in the target population (whole study population that meets inclusion/exclusion criteria) specified in their methods.
- In case of secondary data (e.g., registry, EMR/EHR/EDR, claims, etc.) and authors included all who met inclusion/exclusion criteria (whole study population that meets inclusion/exclusion criteria.)

Mark only one oval.

- Random
- Non-random
- Whole population was invited (no sample was taken) *Skip to question 26*

Sampling, sample size, and selection bias

23. Was a sample size calculation done? *

Yes: if authors mentioned in methods.

No: if sampling was done **but no** sample size calculation was done.

Mark only one oval.

- Yes
- No *Skip to question 26*

Sampling, sample size, and selection bias

24. Did the authors account for ineligibility, non-response, loss to follow-up, change in exposure status over time. etc. in sample size calculation? *

Yes: If authors took a sample and mentioned that they increased the sample size to address any of these issues.

No: If authors took a sample **but did not** mention that they increased the sample size to address any of these issues.

Mark only one oval.

- Yes
- No

25. Was the sample size needed achieved? *

Yes: if authors reached the minimum sample size suggested by sample size the calculator.

No: if authors took a sample and did a sample size calculation **but did not** reach the sample size determined by the calculation.

Mark only one oval.

- Yes
- No

Sampling, sample size, and selection bias

26. Was the baseline selection bias from the investigator accounted for? *

Yes: if authors took a non-random sample (or invited the whole population or took a random sample of participants **but not** everyone agreed to participate) and authors used inverse probability of selection weights.

No: if authors took a non-random sample (or invited the whole population or took a random sample of participants **but not** everyone agreed to participate) and authors **did not** use inverse probability of selection weights.

No baseline selection bias from investigator: if the authors invited the whole population or took a random sample of participants and everyone agreed to participate.

Mark only one oval.

- Yes
- No
- No baseline selection bias from investigator *Skip to question 28*

Sampling, sample size, and selection bias

27. Did the authors compare the distribution of the exposure and outcome between those included (responded) in the study and those not included (did not respond) in the study to check if there was selection bias due to participant' baseline non-response? *

Yes: if the authors checked the distribution (percentage, mean, etc.) of the variables among respondents and non-respondents.

No: if the authors **did not** check the distribution (percentage, mean, etc.) of the variables among respondents and non-respondents.

Mark only one oval.

- Yes
- No

Sampling, sample size, and selection bias

28. Was there a follow-up in the study? *

Yes: if the study was longitudinal (at multiple time points).

No: if the study was cross-sectional (at one point of time).

Mark only one oval.

- Yes
- No *Skip to question 31*

Sampling, sample size, and selection bias

29. Was there loss to follow-up bias? *

If the outcome was measured at multiple time points, then please answer this question based on the main time point of the outcome specified in the methods. If none was specified, then please answer this question based on the last time point.

Yes: if there was follow-up and there was differential loss to follow-up or differential censoring with respect to the exposure.

No: if there was follow-up and there was **no** differential loss to follow-up or **no** differential censoring with respect to the exposure.

Not reported or unknown: if there was a follow-up and authors **did not** report loss to follow-up or censoring within categories of the exposure.

Mark only one oval.

- Yes
- No *Skip to question 31*
- Not reported or unknown *Skip to question 31*

Sampling, sample size, and selection bias

30. Was the loss to follow-up bias accounted for? *

If the outcome was measured at multiple time points, then please answer this question based on the main time point of the outcome specified in the methods. If none was specified, then please answer this question based on the last time point.

Yes: if there was follow-up and there was differential loss to follow-up or differential censoring with respect to the exposure and the authors used inverse probability of censoring weights.

No: if there was follow-up and there was differential loss to follow-up or differential censoring with respect to the exposure and the authors **did not** use inverse probability of censoring weights.

Mark only one oval.

- Yes
- No

Exposure measurement and its validity

31. What was the nature of the exposure? *

Regardless of how the authors actually measured the variable in the study

Subjective: if there aren't any methods to measure the variable besides relying on the self-report of a participant. These include feelings, pain, quality of life, perception, attitude, etc.

Objective: if there are methods to measure the variable besides relying on the self-report of a participant (e.g., through examination or testing). These include knowledge, body examination, scales, blood tests, body measurements, imaging, etc.

Mark only one oval.

- Subjective
- Objective

32. Did the authors use validated measurement for the exposure? *

First yes: if authors conducted internal validation to calculate sensitivity/specificity/PPV/NPV or cited these numbers (no need to go back and looking at the title and content of the cited paper.)

Second yes: if authors stated that they used a validated measurement without sensitivity/specificity/PPV/NPV or conducted validation with experts (no need to go back and looking at the title and content of the cited paper.)

This type of objective measurement does not usually require validation (e.g., surgical intervention): if authors used an objective measurement that does not usually require validation for logistical reasons such as surgical interventions.

No: if authors **did not** do any criterion/content/face validity themselves or **did not** cite any paper/guideline/classification even if the measurement was considered a usual/gold standard measurement.

Mark only one oval.

- Yes, the authors conducted criterion validity or cited the criterion validity from other studies
- Yes, the authors conducted content/face validity or cited the content/face validity from other studies
- This type of objective measurement does not usually require validation (e.g., surgical intervention)
- No

33. Was the measurement bias in the exposure differential or non-differential with respect to the outcome? *

Differential: if exposure data was collected **after** knowing the outcome status (such as in recall bias in case-control or retrospective cohorts and case by case in cross-sectional studies.) This is only when the authors measure the exposure through a primary data collection tool.

Non-differential: if exposure data was collected **before** knowing the outcome status (data was already recorded on the exposure in registry, EMR/EHR/EDR, claims, etc. before ascertaining the outcome.)

Mark only one oval.

- Differential with respect to the outcome
- Non-differential with respect to the outcome

34. Was the measurement bias in the exposure accounted for? *

(e.g., corrected using internal validation or transported sensitivity, specificity, PPV, or NPV)

Yes: if authors corrected exposure using sens/spec/PPV/NPV.

No: if authors **did not** correct exposure using sens/spec/PPV/NPV.

Mark only one oval.

- Yes
- No

Outcome measurement and its validity

35. What was the nature of the outcome? *

Regardless of how the authors actually measured the variable in the study

Subjective: if there aren't any methods to measure the variable besides relying on the self-report of a participant. These include feelings, pain, quality of life, perception, attitude, etc.

Objective: if there are methods to measure the variable besides relying on the self-report of a participant (e.g., through examination or testing). These include knowledge, body examination, scales, blood tests, body measurements, imaging, etc.

Mark only one oval.

- Subjective
- Objective

36. Did the authors use validated measurement for the outcome? *

First yes: if authors conducted internal validation to calculate sensitivity/specificity/PPV/NPV or cited these numbers (no need to go back and looking at the title and content of the cited paper.)

Second yes: if authors stated that they used a validated measurement without sensitivity/specificity/PPV/NPV or conducted validation with experts (no need to go back and looking at the title and content of the cited paper.)

This type of objective measurement does not usually require validation (e.g., surgical intervention): if authors used an objective measurement that does not usually require validation for logistical reasons such as surgical interventions.

No: if authors **did not** do any criterion/content/face validity themselves or **did not** cite any paper/guideline/classification even if the measurement was considered a usual/gold standard measurement.

Mark only one oval.

- Yes, the authors conducted criterion validity or cited the criterion validity from other studies
- Yes, the authors conducted content/face validity or cited the content/face validity from other studies
- This type of objective measurement does not usually require validation (e.g., surgical intervention)
- No

37. Was the measurement bias in the outcome differential or non-differential with respect to the exposure? *

Differential: if the data collector was not blinded to the status of the exposure when measuring the outcome (interviewers bias).

Non-differential: otherwise.

Mark only one oval.

- Differential with respect to the exposure
- Non-differential with respect to the exposure

38. Was the measurement bias in the outcome accounted for? *

(e.g., corrected using internal validation or transported sensitivity, specificity, PPV, or NPV)

Yes: if authors corrected outcome using sens/spec/PPV/NPV.

No: if authors **did not** correct outcome using sens/spec/PPV/NPV.

Mark only one oval.

- Yes
- No

Relation of the exposure measurement bias to the outcome measurement bias

39. Was the exposure measurement bias independent or dependent of the outcome measurement bias? *

Dependent: if both exposure and outcome were measured with errors and both were self-reported or both were ascertained from the same source of data.

Independent: if both were measured with error but one of them was **not** self-reported or both were **not** ascertained from the same source of data.

Mark only one oval.

- Dependent
- Independent

Confounding variables determination and handling in study

40. What was/were the method(s) used for handling baseline confounding? (select all that apply) *

No adjustment for baseline confounding was done: if there was no randomization or quasi-experimental design and the authors did not use any of the methods to adjust for confounding (e.g., only did t-test, chi-square, ANOVA, etc.)

Check all that apply.

- [ ] Randomization
- [ ] Stratification
- [ ] Restriction
- [ ] Matching
- [ ] Regression
- [ ] Propensity score
- [ ] Inverse probability of treatment weighting
- [ ] Standardization
- [ ] Doubly robust methods (inverse probability of treatment weighting + standardization)
- [ ] Instrumental variable analysis besides randomization (difference-in-difference, regression discontinuity design, interrupted time series)
- [ ] No adjustment for baseline confounding was done

41. Did the authors estimate a time-varying effect? *

Yes: if there was a follow-up in the study and authors estimated a time-varying effect.

No:

- if there was a follow-up in the study and authors did not estimate a time-varying effect.
- If there was no follow-up (cross-sectional).

Mark only one oval.

- Yes
- No *Skip to question 43*

Confounding variables determination and handling in study

42. What was/were the method(s) used for handling time-varying confounding? (select all that apply) *

No adjustment for time-varying confounding was done: if there was no randomization or quasi-experimental design and the authors did not use any of the methods to adjust for confounding (e.g., only did t-test, chi-square, ANOVA, etc.)

Check all that apply.

- [ ] Randomization
- [ ] Stratification
- [ ] Restriction
- [ ] Matching
- [ ] Regression
- [ ] Propensity score
- [ ] Inverse probability of treatment weighting using marginal structural models
- [ ] Standardization/parametric g-formula
- [ ] Doubly robust methods (inverse probability of treatment weighting using marginal structural models + standardization)
- [ ] G-estimation
- [ ] No adjustment for time-varying confounding was done

Confounding variables determination and handling in study

43. How did the authors determine the confounding variables that they have adjusted for? (select all that apply) *

Based on Directed Acyclic Graphs (DAGs)/subject matter expertise: if DAG was listed in the paper or cited or stated they used subject matter experts to select confounding variables.

Based on previous literature: if authors mentioned them in the methods and cited previous papers that adjusted for the selected variables.

Based on statistical criteria: if authors used significance testing (or stepwise selection), ML algorithms, or statistical tests.

Based on change of estimate: if authors used 10% change of estimate typically found in epidemiological literature

Based on none of the above: otherwise.

No adjustment for baseline and time-varying confounding was done: if authors did not adjust for any confounding variable in the study.

Check all that apply.

- [ ] Based on Directed Acyclic Graphs (DAGs)/subject matter expertise
- [ ] Based on previous literature
- [ ] Based on statistical criteria
- [ ] Based on change of estimate
- [ ] Based on none of the above
- [ ] No adjustment for baseline and time-varying confounding was done

Missing exposure data

44. Was there any participant with a missing exposure? *

Yes: If authors stated in the text or tables that at least one participant included in the study had a missing value in exposure.

No: If authors reported all categories of a variable and they sum up to the study population.

Not reported or unknown: If authors **did not** report all categories of a variable (e.g., only reported one category of a binary variable and so on) and did not specifically report how many were missing the exposure in text or tables.

Mark only one oval.

- Yes
- No *Skip to question 46*
- Not reported or unknown *Skip to question 46*

Missing exposure data

45. Any methods used for handling missing data in exposure? *

No: if there was missing data in exposure but no imputation was done.

Mark only one oval.

- Yes, missing indicator modeling
- Yes, single imputation with mean value
- Yes, single imputation with predicted value using regression
- Yes, single imputation with last value carried forward
- Yes, multiple imputation
- No

Missing outcome data

46. Was there any participant with a missing outcome? *

If the outcome was measured at multiple time points, then please answer this question based on the main time point of the outcome specified in the methods. If none was specified, then please answer this question based on the last time point.

Yes: If authors stated in the text or tables that at least one participant included in the study had a missing value in outcome.

No: If authors reported all categories of a variable and they sum up to the study population.

Not reported or unknown: If authors **did not** report all categories of a variable (e.g., only reported one category of a binary variable and so on) and did not specifically report how many were missing the outcome in text or tables.

Mark only one oval.

- Yes
- No *Skip to question 48*
- Not reported or unknown *Skip to question 48*

Missing outcome data

47. Any methods used for handling missing data in outcome? *

If the outcome was measured at multiple time points, then please answer this question based on the main time point of the outcome specified in the methods. If none was specified, then please answer this question based on the last time point.

No: if there was missing data in outcome but no imputation was done.

Mark only one oval.

- Yes, missing indicator modeling
- Yes, single imputation with mean value
- Yes, single imputation with predicted value using regression
- Yes, single imputation with last value carried forward
- Yes, multiple imputation
- No

Addressing errors in discussion section

48. Did the authors qualitatively mention the following errors in the discussion section? (select all that apply) *

Please review the whole discussion and not just the limitations section. Only choose an error if it was clearly discussed. If authors vaguely talked about variables and mentioned that they could impact the results without how pointing the type of error it can lead to, please do not select an error based on your interpretation.

Check all that apply.

- [ ] Random error
- [ ] Selection bias
- [ ] Measurement bias
- [ ] Confounding bias
- [ ] Missing data
- [ ] None of the above

Comments

49. Do you have any comments?

This is where you add any comments on the paper (e.g., you have an issue with one of the questions, you explain why you recuse yourself from reviewing this paper, etc.)

If you have more than one comment, please separate them with a semicolon.

50. What is/are the comment/comments related to?

Check all that apply.

- [ ] Recusal from reviewing the paper
- [ ] Study design
- [ ] Study population
- [ ] Sampling
- [ ] Selection bias
- [ ] Exposure measurement bias
- [ ] Outcome measurement bias
- [ ] Confounding bias
- [ ] Exposure missing data
- [ ] Outcome missing data
- [ ] Error discussion
- [ ] Conflating task